

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Pseudocode Writing Task

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO5 – Naturalization (independent application using OpenCV)P5	Assessment:	Assessment Tool 2 – Pseudocode Writing Task
Weightage:	25% of CIA	Total Marks:	100
CO4 Statement:	Develop computer vision applications using OpenCV for performing recognition and analysis on images and videos. (BTL6)		
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Section / Batch:	Third year / B.Tech IT	Date:	26/08/2026

Code of Conduct

I KAVIYA V / 192421022 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Kaviya V

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation	
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach	
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs	
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear	
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient	

QUESTION:

6. Real-Time Object Detection Pipeline

A system must:

- A. Capture video frames
- B. Detect objects using a trained model
- C. Draw bounding boxes and labels
- D. Display results in real time

Constraints:

- Must process frames efficiently
- Handle multiple objects

Write detailed pseudocode with modular steps for detection, visualization, and efficient looping.

ANSWER:

PSEUDOCODE:

BEGIN

FUNCTION load_model()

 Load trained object detection model

 RETURN model

END FUNCTION

FUNCTION capture_frame(camera)

 Read frame from camera

 IF frame is not available THEN

 RETURN NULL

 END IF

 RETURN frame

END FUNCTION

FUNCTION detect_objects(model, frame)

 Resize frame to required size

 Send frame to model

 Get:

 Object classes

 Confidence scores

 Bounding boxes

 RETURN detections

END FUNCTION

```
FUNCTION visualize(frame, detections)

  FOR each detection IN detections

    IF confidence > threshold THEN

      Draw bounding box

      Draw object class label

      Draw confidence score

    END IF

  END FOR

  RETURN frame

END FUNCTION

FUNCTION main()

  model = load_model()

  camera = OpenCamera()

  WHILE camera is running

    frame = capture_frame(camera)

    IF frame is NULL THEN
      BREAK
    END IF

    detections = detect_objects(model, frame)

    result = visualize(frame, detections)

    Display result

    IF user presses "Q" THEN
      BREAK
    END IF

  END WHILE

  Release camera

  Close display window
```

END FUNCTION

CALL main()

END

MODULAR STRUCTURE:

MAIN()

